Deep Learning-Based Three-Phase Voltage Parameter Estimation

扰动条件下的三相电压参数估计

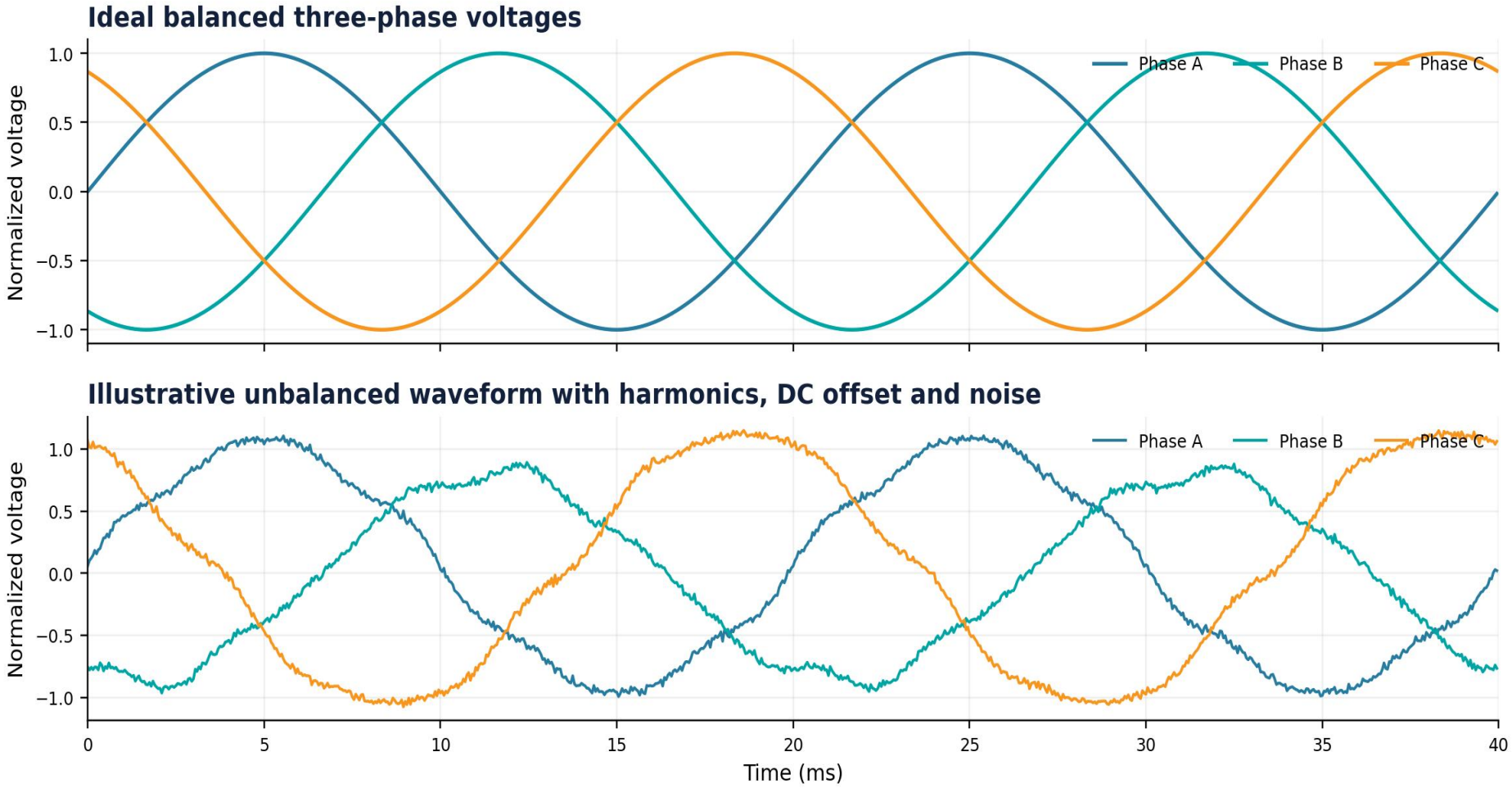
Under Disturbances: A Study of Sampling Frequency and Observation Window | 采样频率与观测窗口研究

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CURRENT FINDINGS + REVISED EXPERIMENT PLAN

IDEAL/REALISTIC THREE PHASE SIGNAL | 从理想三相电压到真实波形



IDEAL:
Equal Amplitudes + 120° Separation

理想情况：

三相电压幅值相等，相位彼此相差 120°。

REALISTIC:
Unbalance + Harmonics + DC + Noise

实际情况：

电压不平衡、谐波、直流偏置及噪声。

The Estimation Trade-Off | 估计精度与成本的权衡

CORE PROBLEM | 核心问题

How can a system estimate amplitude, phase angle and fundamental frequency accurately, while using as little observation time and computation as possible?

如何在尽可能缩短观测时间、降低计算开销的同时，准确估计系统的幅值、相位角和基波频率？

RESEARCH QUESTION | 研究问题

What combination of sampling frequency, observation window, dataset size and network design gives an acceptable accuracy-cost balance under waveform disturbances?

在波形扰动条件下，采样频率、观测窗口、数据集规模和网络设计如何组合，才能在估计精度与计算成本之间取得可接受的平衡？

Objectives | 研究目标

01

Estimate 7 targets

Three fundamental amplitudes, three phase angles and one shared frequency.

三个基波幅值、三个相位角，以及一个三相共用的频率。

02

Study acquisition settings

Vary sampling frequency, observation window and training-data quantity.

改变采样频率、观测窗口长度和训练数据量。

Design principle | 设计原则

Estimation error, observation delay, sampling constraints and computation must be considered together.

必须综合考虑估计误差、观测延迟、采样约束和计算开销。

Mathematical Model | 数学模型

$$v_p(t) = A_p \sin(2\pi ft + \varphi_p) + \Sigma_h A_{h,p} \sin(2\pi hft + \varphi_{h,p}) + D_p + n_p(t)$$

$(v_p(t))$	Instantaneous phase voltage	各相瞬时电压
(p)	Phase A, B, or C	A、B或C相
(t)	Time (s)	时间（秒）
(A_p)	Fundamental peak amplitude	基波峰值幅度
(f)	Fundamental frequency (Hz)	基波频率（Hz）
(φ_p)	Fundamental phase angle	基波相位角
(h)	Harmonic order	谐波次数
$(A_{h,p})$	Harmonic peak amplitude	谐波峰值幅度
$(\varphi_{h,p})$	Harmonic phase angle	谐波相位角
$(\sum_h)$	Sum of harmonic components	各次谐波分量之和
(D_p)	DC offset	直流偏置
$(n_p(t))$	Time-varying noise	时变噪声

Sampling Frequency and Observation Window Define the Input | 采样频率与观测窗口决定输入

The sampled waveform length determines both how much signal information is available and the neural network input size.

采样波形的长度决定了可获取的信号信息量以及神经网络的输入维度。

Sampling equations | 采样公式

$$T_{obs} = 1 / f$$

$$N = \text{ceil}\,(f_s \cdot T_{obs})$$

$$x = [v_a[0{:}N-1],\ v_\beta[0{:}N-1],\ v_c[0{:}N-1]]$$

Input dimension = 3N features | 输入维度 = 3N 个特征

Supervised learning maps waveform samples to known parameters
从波形样本到参数的监督学习

1

Signal synthesis | 信号生成

Generate A, φ, f and disturbance labels.

2

Sampling + window | 采样与观测窗口

Create three sampled phase arrays.

3

Feature vector x | 特征向量 x

Concatenate A, B and C samples.

4

Train network | 训练网络

Compare predictions with known labels.

5

Evaluate | 评估

MAE, RMSE and cost on unseen data.

Network input | 网络输入

$$x \in \mathbb{R}^{3N}$$

Three-phase voltage samples | 三相电压采样值
(e.g., 609 features for 203 samples/phase | 例如，每相采集 203 个采样点时，输入向量包含 609 个特征。)

Target output | 目标输出

$$y = [A_a, A_\beta, A_c, \varphi_a, \varphi_\beta, \varphi_c, f]^T$$

Seven continuous outputs -> a regression problem, not classification. | 七个连续数值输出，因此属于回归任务

Data split | 数据集划分

70% training updates weights/biases; 15% validation selects the model; 15% held-out test evaluates final performance.

70% 用于训练并更新权重和偏置，15% 用于验证并选择模型，剩余 15% 用于独立测试，以评估模型的最终性能

Simulation settings and disturbance coverage | 仿真设置与扰动范围

Several ranges are explicit experimental assumptions rather than universal standards. | 表中所列的若干参数范围是本实验明确采用的假设，并非普遍适用的标准。

Fundamental frequency | 基波频率	50 Hz nominal; 49.5-50.5 Hz
Nominal phase voltage | 标称相电压	220 V RMS ≈ 311.13 V peak
Sampling frequency | 采样频率	1, 2, 5, 10 kHz
Observation window | 观测窗口	2, 5, 10, 15, 20 ms
Harmonics | 谐波	5th and 7th
Noise / DC offset | 噪声 / 直流偏置	Moderate: SNR 30-40 dB, DC up to 2%; severe: 20-30 dB, DC up to 5%
Dataset split | 数据集划分	70% / 15% / 15%

INTERPRETATION | These are simulated results; validation on measured grid voltage remains future work.

结果说明 | 这些结果来自仿真；利用实际测量的电网电压进行验证仍是后续工作。

Supervised neural network | 监督神经网络

A fully connected feedforward network learns to estimate seven continuous voltage parameters.
采用全连接前馈神经网络，学习估计三相电压的七个连续参数。

INPUT
输入层
3N
Voltage samples
电压采样值

BOTTLENECK
瓶颈层
36
Linear layer
线性层

HIDDEN LAYER 1
隐藏层 1
128
tanh activation
tanh 激活函数

HIDDEN LAYER 2
隐藏层 2
64
ReLU activation
ReLU 激活函数

OUTPUT LAYER
输出层
7
Linear regression outputs
线性回归输出

Input size varies with sampling frequency and observation window. | 输入维度随采样频率和观测窗口而变化。

PREDICTION TARGET | 预测目标

$$[A_a, A_\beta, A_c \ , \varphi_a, \varphi_\beta, \varphi_c \ , f]^T$$

3 amplitudes + 3 phase angles + 1 frequency

3 个幅值 + 3 个相位角 + 1 个频率

TRAINING SETUP | 训练设置

70%	15%	15%

Train / Validation / Test

训练集 / 验证集 / 测试集
Adam optimizer (where specified) | 按实验设置使用 Adam 优化器

EVALUATION | 性能评估

MAE
平均绝对误差

RMSE
均方根误差

Evaluate final performance on held-out test data.
使用独立测试集评估最终性能。

5 | EXPERIMENTS/RESULTS | 结果

Sampling Frequency: More Samples Generally Improved Accuracy | 采样频率对精度的影响

Sampling Frequency	Dataset Size	Mean Amp MAE (V)	Mean Phase MAE (°)	Freq MAE (Hz)	Training Time (s)
1 kHz	2000	1.82	0.47	0.1	54.9
1 kHz	20000	1.64	0.43	0.09	366.1
2 kHz	20000	1.24	0.33	0.07	413.2
5 kHz	20000	0.92	0.23	0.04	416.2
10 kHz	2000	0.98	0.2	0.04	86.4
10 kHz	20000	1.1	0.23	0.04	264
5 kHz (Best Compromise)	5000	1.13	0.24	0.05	136
5 kHz (Best Compromise)	10000	1.03	0.24	0.04	242.3

Finding | 主要发现

10 kHz gives the lowest reported errors at this data size, but increases the number of features from 300 at 5 kHz to 600 at 10 kHz for 20 ms.

在该数据集规模下，10 kHz 采样频率的已报告误差最低，但对于 20 ms 的观测窗口，输入特征数量也从 5 kHz 时的 300 个增加到 10 kHz 时的 600 个。

Observation Window: Shorter windows increased the error | 观测窗口：窗口较短时误差较大

Window (ms)	Dataset Size	Mean Amp MAE (V)	Mean Phase MAE (°)	Freq MAE (Hz)	Training Time (s)
2 ms	20000	13.96	2.42	0.26	560.5
5 ms	20000	9	1.62	0.25	1027.3
10 ms	10000	2.11	0.5	0.23	427.9
15 ms	5000	1.02	0.29	0.09	280.3
20 ms	20000	1.52	0.26	0.04	319.2
20 ms	2000	105	0.21	0.04	91.9
20 ms (Strongest Overall)	10000	0.92	0.2	0.04	272.3
15 ms (Faster Compromise)	5000	1.02	0.29	0.09	280.3

Takeaway | 结论要点

A longer window gives the network more waveform information, especially for frequency estimation.

较长的观测窗口能够为神经网络提供更多的波形信息，尤其有助于提高频率估计的准确性。

The best setting depends on the accuracy-cost priority | 最佳设置取决于精度与成本优先级

The reported study suggests a practical compromise, but these numerical comparisons must be rerun under the corrected signal/window design.

已报告的研究结果表明，该方案在实际应用中提供了一种可行的折中选择，但仍需在修正后的信号模型和观测窗口设计下重新进行这些数值比较。

Configuration	Amp MAE / RMSE	Phase MAE / RMSE	Freq MAE / RMSE	Train time
5 kHz | 20 ms | 10k	0.98 / 1.35 V	0.25 / 0.35°	0.05 / 0.07 Hz	166.6 s
10 kHz | 20 ms | 10k	0.83 / 1.10 V	0.21 / 0.29°	0.04 / 0.05 Hz	234.0 s
10 kHz | 20 ms | 20k	0.74 / 1.01 V	0.21 / 0.29°	0.04 / 0.05 Hz	363.2 s

Balanced option | 平衡方案

5 kHz, 20 ms, 10k examples halves the sample-rate input dimension relative to 10 kHz and trained faster in this run.

在本次实验中，采用 5 kHz 采样频率、20 ms 观测窗口和 10,000 个样本的数据集，相比 10 kHz 采样频率，将输入特征维度减半，且训练速度更快。

Lowest-error reported option | 已报告的最低误差方案

10 kHz, 20 ms, 20k examples achieved the lowest errors among these settings, with a higher training cost.

在这些实验设置中，采用 10 kHz 采样频率、20 ms 观测窗口和 20,000 个样本的数据集，获得了最低的估计误差，但训练成本更高。

Dataset-size lesson | 数据量启示

More training examples did not improve every experiment consistently. The impact depends on settings and model training.

增加训练样本数量并未在所有实验中持续提升性能，其影响取决于实验设置和模型训练情况。

Important: training time is not per-signal inference time; actual deployment latency has not been measured.

What the current results support - and what they do not | 当前结果能说明什么、尚不能说明什么

The current results are evidence from a simulation study, not proof of real-time or real-grid performance. | 目前的结果是仿真研究的证据，尚不能证明该方法在实时运行或实际电网中的性能。

Supported by the report | 报告支持的结论

Longer observation windows and higher sampling frequencies generally reduced estimation errors.

较长的观测窗口和较高的采样频率通常能够降低参数估计误差。

Promising balance | 有希望的平衡方案

5 kHz, 20 ms and 10,000 examples is a useful candidate for balancing error, feature count and training time.

采用 5 kHz 采样频率、20 ms 观测窗口和 10,000 个样本的数据集，是一种兼顾估计误差、输入特征数量和训练时间的可行方案。

Not established yet | 尚未验证

Performance under the 13th/17th harmonics, the corrected ≥20.3 ms window, real measured waveforms and edge-device latency.

仍需进一步验证模型在包含 13 次和 17 次谐波、采用修正后的至少 20.3 ms 观测窗口、使用实际测量波形以及在边缘设备上运行时的性能与推理延迟。

Conclusion | 结论

Deep supervised learning is promising for parameter estimation, but model choice should follow controlled validation under a signal model that satisfies the sampling constraints.

深度监督学习在参数估计方面具有良好的应用前景，但模型的选择应基于受控验证，并确保所采用的信号模型满足采样约束。

Next actions before finalizing the report | 定稿前的后续工作

01	**Sampling compliance**	Revise full-cycle window to at least 20.202 ms; use 20.3 ms at 10 kHz. | 将完整周期观测窗口修正为至少 20.202 ms；在 10 kHz 采样频率下，采用 20.3 ms。
02	**Harmonics + Nyquist**	Add 13th and 17th harmonics. At 50.5 Hz, the 17th is 858.5 Hz; theoretical fs must exceed 1.717 kHz. | 加入第 13 次和第 17 次谐波。在 50.5 Hz 时，第 17 次谐波的频率为 858.5 Hz；理论上，采样频率必须高于 1.717 kHz。
03	**Activation ablation**	Compare ReLU/ReLU with tanh/ReLU under identical training settings. | 在相同的训练设置下，对比 ReLU/ReLU 与 tanh/ReLU 两种激活函数组合。
04	**Layer-width trade-off**	Compare 32-16, 64-32 and 128-64 hidden widths; select by validation error and complexity. | 比较 32-16、64-32 和 128-64 三种隐藏层神经元配置，根据验证误差和模型复杂度选择合适的结构。
05	**Report revised results**	Rerun the main grid; preserve a separate test set and clearly label old results vs updated results. | 重新运行主要实验组合，保留独立测试集，并明确区分旧实验结果与更新后的实验结果。

Selected references | 主要参考文献

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Thank you | 谢谢